

# SREE GNANESH DOMMETI

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## Education

### University of Maryland, College Park

*M.S. in Applied Machine Learning; GPA: 3.9 / 4.0*

May 2026

College Park, MD

### National Institute of Technology, Calicut

*B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10*

May 2022

Calicut, India

## Experience

### FROOT Lab, University of Maryland

November 2025 – Present

*Research Assistant, Advisor: Alan Zaoring Liu*

College Park, MD

- Contributing to ProjectASAP, an approximation-first telemetry research effort that serves approximate analytics from streaming sketches instead of repeated raw time-series scans.
- Built PromQL workload-analysis/sketch-planning paths and NetArena MALT benchmark tooling with Dockerized LiteLLM prompts, output validation, retries, and local submissions.

### GAMMA Lab, University of Maryland

January 2026 – May 2026

*Research Assistant, Advisor: Dinesh Manocha*

College Park, MD

- Contributed to ongoing grounded MLLM/video reasoning evaluation tooling that scores intermediate reasoning steps against visual/textual evidence.

### Qualcomm

July 2022 – July 2024

*Software Engineer*

Hyderabad, India

- Owned LTE Layer 3 RRC Cell Selection and Security modules for 4G/5G, CAT-M1, and NB-IoT modem stacks.
- Built Python log-analysis, regression-tracking, changelist-routing, and fuzzing automation over modem traces, reducing manual triage and assignment effort by 40%.

### Connycr

June 2025 – August 2025

*ML Engineer Intern, CampusAI Event Recommendation*

College Park, MD

- Built a personalized campus event recommendation system with FastAPI, Elasticsearch lexical retrieval, MiniLM semantic search, and reciprocal-rank fusion; achieved sub-300 ms p95 retrieval and sub-600 ms p95 end-to-end latency.
- Implemented ALS/BPR collaborative filtering and offline ranking evaluation over implicit interactions, tracking Recall@K, NDCG@K, MRR@K, cold-start recall, long-tail coverage, and RSVP/save-rate proxy metrics with MLflow.

## Projects

### Benchmarking LLM Inference Frameworks | *Python, PyTorch, vLLM, SGLang, Docker, AWS EC2 GPU*

- Deployed vLLM and SGLang to serve Mistral-7B-Instruct-AWQ on EC2 GPU; benchmarked 1–64 concurrent users for latency, throughput, cost, GPU utilization, and saturation behavior.

### CallIntel Agent - Audio Support-Call QA | *Python, GCP, AWS Bedrock, OpenTelemetry, Datadog*

- Built an audio-to-RAG support QA prototype with PII redaction, cited answers, Gemini/OpenAI/Claude/Bedrock payload validation, GCP artifacts, and Datadog-compatible OpenTelemetry quality/latency metrics.

### Real-Time Fraud Detection ML Lifecycle | *PyTorch, Airflow, dbt/Snowflake, Databricks, AWS ECS*

- Extended a fraud classifier into an ML lifecycle pipeline with DVC/ClearML tracking, drift reports, promotion gates, Dockerized FastAPI inference, Airflow orchestration, dbt/Snowflake marts, and Databricks job specs.

## Skills

**Languages:** Python, SQL, C, C++, Rust

**ML / GenAI:** PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace, XGBoost, LightGBM, LangChain, LangGraph, LlamaIndex, RAG, Gemini JSON APIs, AWS Bedrock payloads, vLLM, SGLang

**Distributed / Serving:** Accelerate, Ray, model-serving benchmarks, GPU inference profiling, activation checkpointing, vLLM/SGLang runtime comparison

**Data / Infra:** Kafka, Arroyo, Spark, Databricks/Spark job specs, Snowflake/dbt artifacts, Redis, PostgreSQL, Elasticsearch, FAISS, Pinecone, Prometheus, Grafana, OpenTelemetry/Datadog-compatible OTLP

**MLOps:** Docker, Kubernetes, Airflow DAGs, AWS ECS/EC2/S3/SageMaker/Lambda, GCP Cloud Run/Pub/Sub/BigQuery/IAM, Terraform, MLflow, ClearML, DVC, W&B, GitHub Actions, EvidentlyAI